

LD1-A Sparse CIC and Canonical-Convolution Reference Specification

Deterministic flat-node aggregation, sparse stencil scatter, normalization, and HDR diagnostics

mdstats development specification

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Status and scope

This specification implements architecture gate **LD1-A** for `mdstats 0.19.45a0`. It is subordinate to the *Dynamical Framework and Density Plotting Architecture Standard*.

The gate introduces a deliberately simple sparse numerical oracle:

weighted periodic samples → sorted CIC node masses → sparse canonical stencil scatter → normalized flat-node field .

The representation stores sorted logical flat indices and positive values. It is not the production block-sparse backend. Block packing, partial-block masks, transactional sparse storage planning, and public backend selection belong to LD1-B and LD4.

This gate does not change the default dense backend, the default `legacy_spectral_v1` operator, scene composition, probability-shell meshing, or framework-edge quadrature.

Scientific ownership and citations

Periodic trilinear cloud-in-cell assignment follows the particle-mesh construction of Hockney and Eastwood (1988). Highest-density-region thresholds follow Hyndman (1996). The finite-support `discrete_periodized_v1` stencil was specified and implemented in LD0-K. The sparse flat-node representation, deterministic accumulation order, resource policy, residual normalization, and debugging conversion limits are project-specific `mdstats` definitions.

Inputs

Weighted samples

The only source record is the existing immutable `PeriodicWeightedSamples3D`:

```
PeriodicWeightedSamples3D(  
    fractional_positions: float64[n_samples, 3], # folded to [0, 1)
```

```

weights: float64[n_samples],          # nonnegative
source_provenance: DensitySourceProvenance,
total_measure: float,
measure_kind: "occupancy" | "arc_length",
measure_units: str,
sample_group_ids: int64[n_samples] | None,
)

```

The weights must satisfy

$$\left| \sum_s w_s - M \right| \leq 5 \times 10^{-13} \max(1, M),$$

where M is `total_measure`.

Grid and cell

The logical shape is

$$\mathbf{N} = (N_1, N_2, N_3), \quad N_i \in \mathbb{N}^+.$$

Logical node (i, j, k) lies at

$$\mathbf{f}_{ijk} = \left(\frac{i}{N_1}, \frac{j}{N_2}, \frac{k}{N_3} \right).$$

The row-vector display cell $H \in \mathbb{R}^{3 \times 3}$ must be finite and nonsingular. Its voxel volume is

$$\Delta V = \frac{|\det H|}{N_1 N_2 N_3}.$$

Canonical kernel

LD1-A supports only `discrete_periodized_v1`. The Gaussian bandwidth obeys $\sigma \geq 0$ and the kernel-tail tolerance lies in $[10^{-15}, 10^{-3}]$.

Sparse-only canonical stencil support

LD0-K stores both a dense logical stencil and its active support. LD1-A adds `PeriodicGaussianStencilSupport`, which stores only:

```

PeriodicGaussianStencilSupport(
    grid_shape,
    display_cell,
    gaussian_bandwidth,
    kernel_tail_tolerance,
    cutoff_radius,
    active_flat_indices,    # sorted, unique int64
    active_weights,         # positive float64, sum exactly one within tolerance
    pre_normalization_sum,
    normalization_factor,
    periodic_image_contribution_count,
    covariance,
    metadata,
)

```

Candidate integer images are enumerated using the reciprocal-plane bounds already specified by LD0-K. Retained image contributions are mapped to canonical periodic flat indices and accumulated in image-enumeration order. The final origin weight receives the residual correction

$$g_0 \leftarrow g_0 + \left(1 - \sum_{\delta} g_{\delta} \right).$$

No logical dense stencil is allocated. A guarded `to_dense_values(max_nodes=...)` helper exists only for bounded testing and debugging.

For $\sigma = 0$, the support is the exact identity:

$$\mathcal{S} = \{0\}, \quad g_0 = 1.$$

Deterministic sparse CIC aggregation

For one folded sample $\mathbf{f}_s$, define

$$\mathbf{y}_s = \mathbf{N} \odot \mathbf{f}_s, \quad \mathbf{i}_s = \lfloor \mathbf{y}_s \rfloor, \quad \mathbf{d}_s = \mathbf{y}_s - \mathbf{i}_s.$$

For $\epsilon \in \{0, 1\}^3$, the periodic target node is

$$\mathbf{n}_{s\epsilon} = (\mathbf{i}_s + \epsilon) \bmod \mathbf{N},$$

with contribution

$$m_{s\epsilon} = w_s \prod_{a=1}^3 \begin{cases} 1 - d_{sa}, & \epsilon_a = 0, \\ d_{sa}, & \epsilon_a = 1. \end{cases}$$

Contributions are generated in the fixed order

(ox, oy, oz) lexicographic, then original sample order

and accumulated with unbuffered float64 additions into sorted unique flat nodes. Zero contributions are omitted. The deposited measure must satisfy

$$\left| \sum_j m_j - M \right| \leq 5 \times 10^{-13} \max(1, M).$$

The output is `SparseCICNodeMasses3D`.

Sparse canonical convolution

Let occupied CIC nodes be (j, m_j) and canonical stencil support be (δ, g_δ) . Every pair contributes

$$\tilde{m}_{(j+\delta) \bmod \mathbf{N}} + = m_j g_\delta.$$

Pairs are generated in the fixed order

stencil flat-index order, then occupied CIC flat-index order

so the addition order tracks the dense direct canonical convolution. Periodic target indices are sorted only for storage; additions retain the declared pair order.

If the raw scattered measure is M_{raw} , apply

$$\beta = \frac{M}{M_{\text{raw}}}, \quad m_g^{\text{norm}} = \beta \tilde{m}_g.$$

A deterministic residual correction is applied to the first stored node. Density is then

$$\rho_g = \frac{m_g^{\text{norm}}}{\Delta V}.$$

The output is `SparseCanonicalDensityReference3D`, a flat-node reference field that implements the backend-neutral scalar-field and periodic-node-access contracts.

Highest-density-region details

For $0 < q < 1$, sort positive stored densities in descending order and choose the first threshold c_q satisfying

$$\Delta V \sum_{\rho_g \geq c_q} \rho_g \geq qM.$$

All exact ties at c_q are included. `SparseHDRDetails` records:

- requested mass fraction;
- threshold;
- achieved mass fraction;
- selected node count;
- exact threshold-tie count;
- selected and total measures.

Implicit zero nodes do not alter a positive threshold for $q < 1$.

Public records and functions

```
build_periodic_gaussian_stencil_support(...)
aggregate_periodic_cic_sparse(...)
scatter_periodic_stencil_sparse(...)
prepare_sparse_canonical_density_reference(...)
```

The main records are:

```
PeriodicGaussianStencilSupport
SparseCICNodeMasses3D
SparseHDRDetails
SparseCanonicalDensityReference3D
```

All public arrays are C-contiguous defensive copies, read-only, shape-validated, and finite. Sparse flat indices are strictly increasing and unique.

Resource and failure policy

The reference path uses explicit hard limits:

```
max_cic_contributions
max_stencil_candidate_contributions
max_kernel_pairs
max_workspace_bytes
max_nodes for debugging conversion
```

Before allocating contribution or pair vectors, the implementation computes a conservative workspace bound. A request exceeding any limit raises `GraphComplexityError` before the corresponding large allocation.

The reference path may allocate arrays proportional to the CIC contribution count or kernel-pair count. It is therefore a correctness oracle, not the final scalable backend. LD1-B must replace flat pair materialization with block-oriented production storage while retaining LD1-A as a test oracle.

Error cases

The implementation rejects:

1. nonfinite or nonfolded weighted samples;
2. negative weights or inconsistent total measure;
3. invalid logical shapes or singular cells;
4. noncanonical smoothing operators;
5. nonpositive sparse stored masses or densities;
6. mismatched CIC and stencil shapes;
7. resource-limit violations;
8. dense debugging conversion beyond `max_nodes`;
9. zero or destroyed measure after deposition or convolution.

Validation matrix

Required fixtures are:

Case	Required property
orthogonal off-grid samples	dense-direct agreement
LTA primitive cell	exact Cartesian metric and dense-direct agreement
face crossing	periodic continuity and dense-direct agreement
edge crossing	periodic continuity and dense-direct agreement
corner crossing	periodic continuity and dense-direct agreement
multiple images in support	canonical image aggregation
overlapping sources	deterministic duplicate aggregation
bimodal hopping	multimodal field and HDR agreement
independent ensemble	arbitrary nonnegative weights
$\sigma = 0$	exact identity convolution

Acceptance against dense direct convolution of `discrete_periodized_v1` is:

relative L1 field error	$\leq 2e-11$
relative L-infinity field error	$\leq 5e-11$
absolute integral error	$\leq 5e-13 * \max(1, \text{total_measure})$
HDR threshold absolute difference	$\leq 5e-12 * \max(1, \text{reference maximum})$
achieved HDR mass-fraction difference	$\leq 5e-13$

Additional gates require:

- repeated identical inputs produce byte-identical sparse indices, values, and metadata;
- periodic integer translations agree within the field tolerances;
- sparse CIC agrees with dense CIC within 2×10^{-16} absolute node mass;
- sparse-only stencil support agrees with the LD0-K dense stencil within 2×10^{-16} absolute weight;
- every resource limit is exercised by a focused failure test;
- the unchanged dense legacy path remains compatible with `mdstats 0.19.44a0`.

Non-objectives and next gate

LD1-A does not expose `grid_backend="local_sparse"` through production plotting options. It does not pack blocks, serialize block fields, render sparse clouds, or extract sparse meshes.

The next gate is **LD1-B**, which will implement production atomic block packing, partial-block masks, transactional sparse preflight, public node access, structured serialization, and the localized LTA storage benchmark.

References

1. Hockney, R. W., and J. W. Eastwood. *Computer Simulation Using Particles*. Taylor & Francis, 1988. Periodic particle-mesh cloud-in-cell assignment.
2. Hyndman, R. J. "Computing and Graphing Highest Density Regions." *The American Statistician* **50** (1996): 120-126. DOI: 10.1080/00031305.1996.10474359.